



DEVELOPMENT OF THE MUSCULOSKELETAL SYSTEM

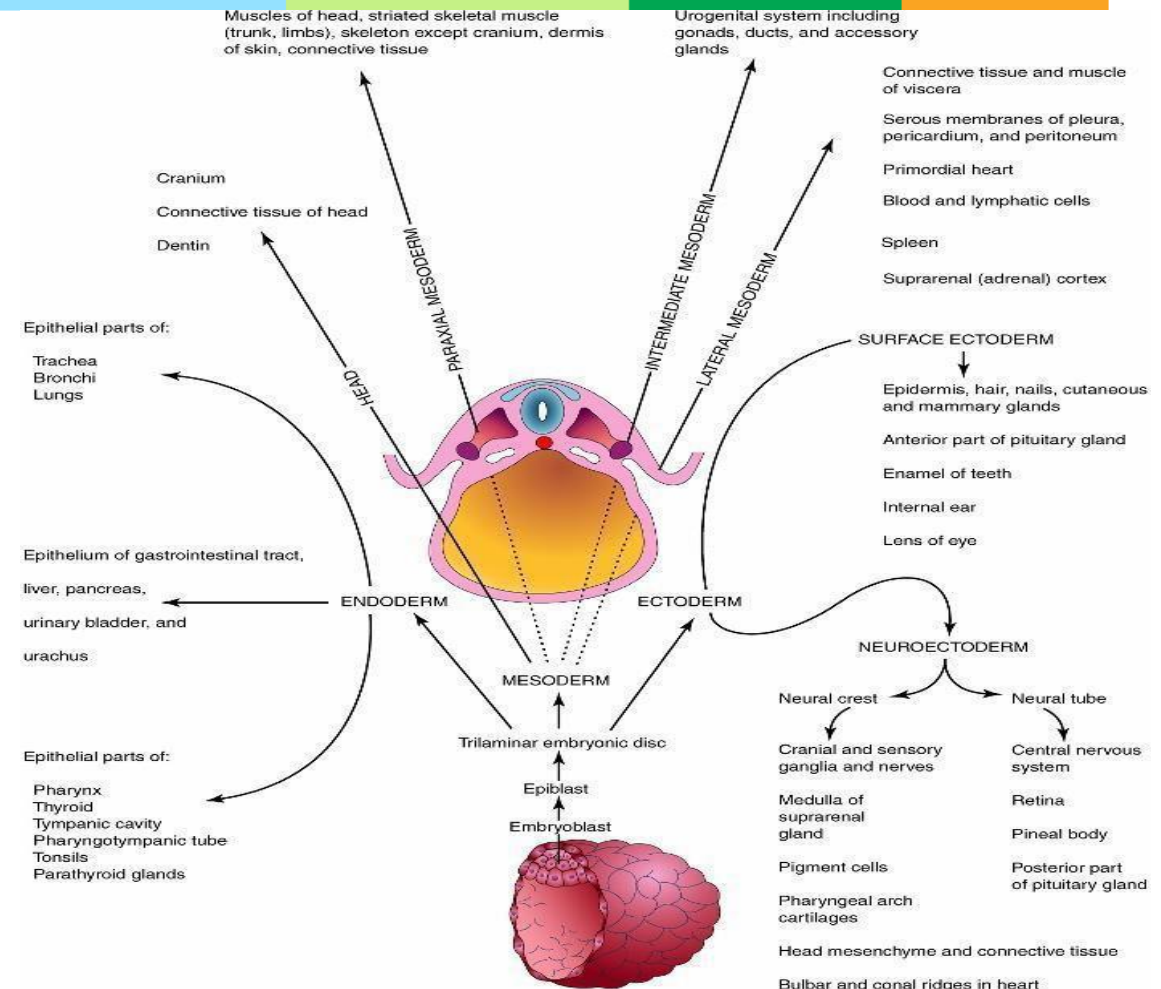
LEARNING OBJECTIVES

1. Indicate the origin of skeletal, smooth, and cardiac muscle.
2. Briefly describe the differentiation of myotomal cells into muscle fibers.
3. Describe the differentiation of myotomes into epimere and hypomere and give their corresponding nerve supply.
4. Give examples of muscles derived from epimere and hypomere in different parts of the neck and trunk.
5. Briefly describe the origin of the head and limb musculature.
6. Briefly describe the origin of the skeletal system.

MUSCULAR SYSTEM DEVELOPMENT

By day 17, the loosely arranged intraembryonic mesoderm becomes differentiated into 3 distinct regions:

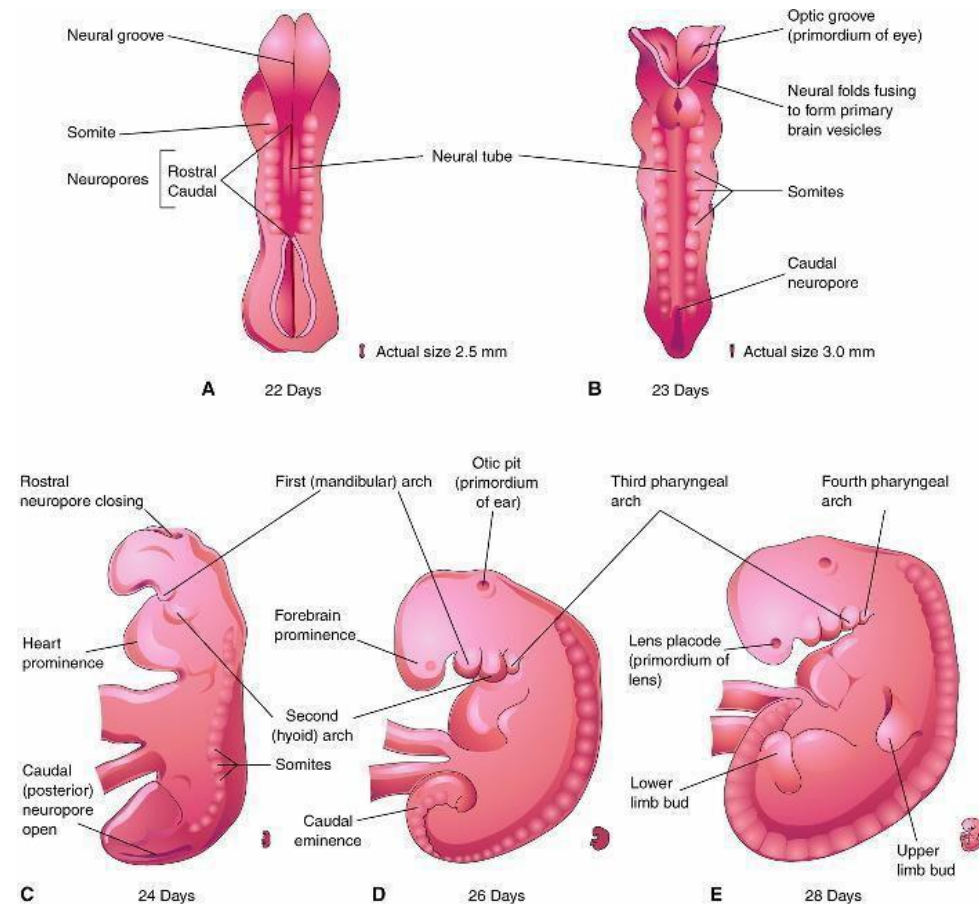
1. Paraxial mesoderm
2. Intermediate mesoderm
3. Lateral plate mesoderm Splanchnic (visceral) layer Somatic (parietal) layer



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DEVELOPMENT OF MUSCULAR SYSTEM

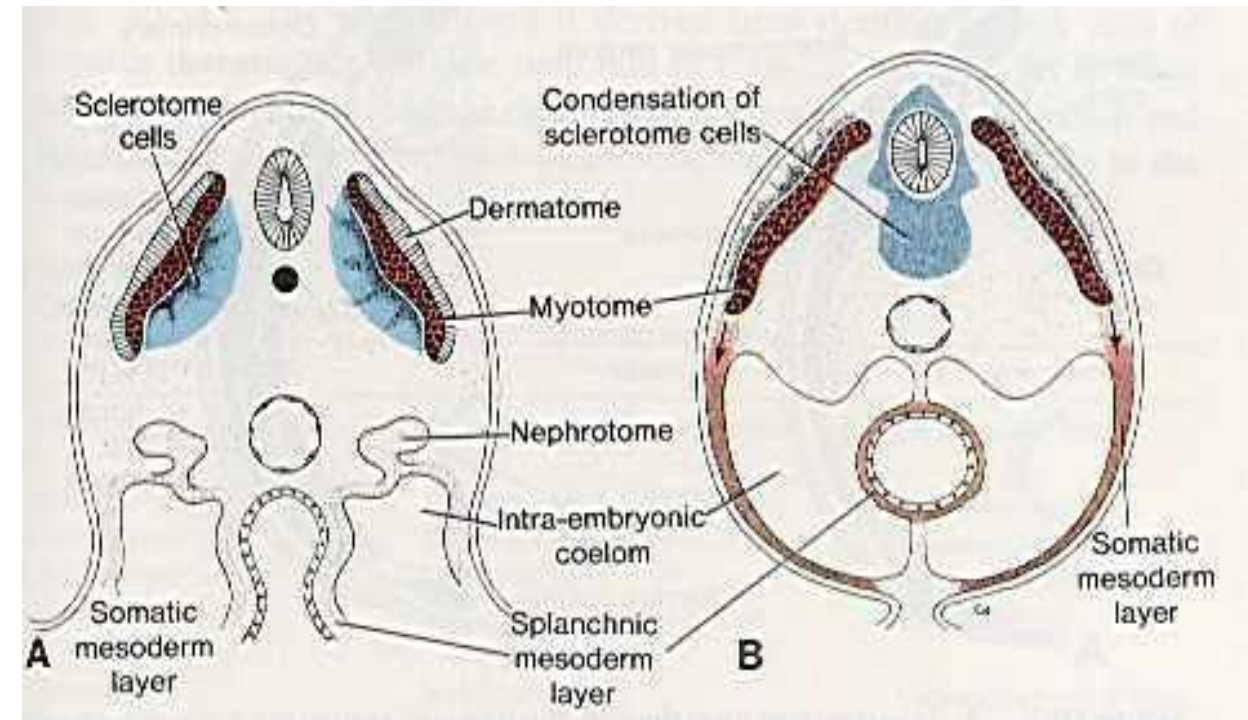
- On day 20, the paraxial mesoderm begins to divide into segmental blocks called somites.
- The first pair of somites appear in the cervical region
- Subsequent pairs appear in a craniocaudal sequence, approximately 3 pairs per day, until 42-44 pairs are present by the end of the fifth week



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MUSCULAR SYSTEM DEVELOPMENT

- Beginning in the fourth week, each somites differentiate into a ventromedial sclerotome and a dorsolateral dermomyotome. The latter further differentiates into dermatome and myotome.
- Sclerotome (“sclero” - hard) - bones, cartilage, ligaments of the vertebral column and parts of the base of the skull
- Dermatome (“derma” - skin) - dermis and hypodermis
- Myotome (“myo” - muscle) - skeletal muscles

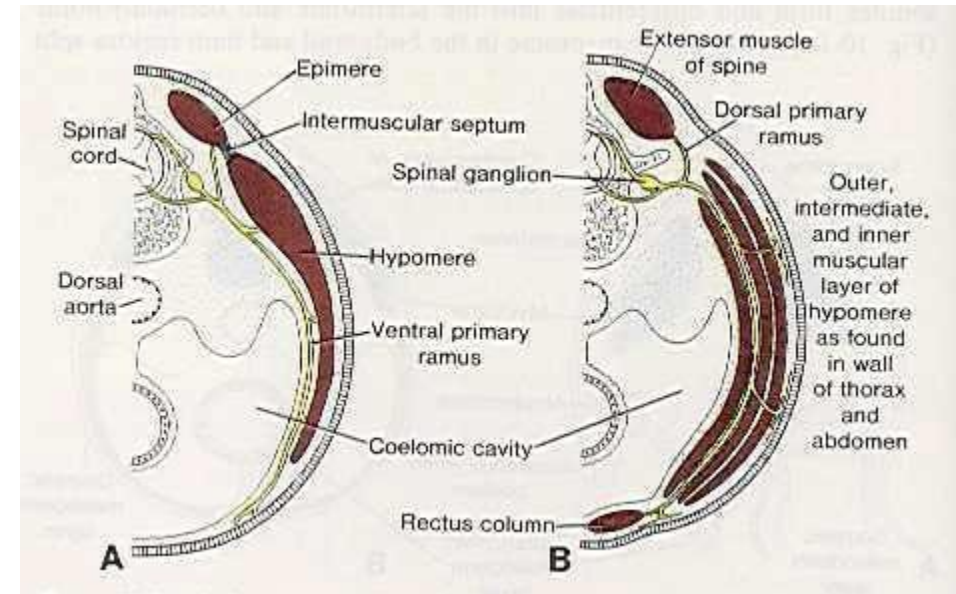


Lateral plate mesoderm - splanchnic - most smooth muscles of visceral organs, cardiac muscle

MUSCULAR SYSTEM DEVELOPMENT

SKELETAL MUSCLES

By the end of the fifth week, body wall musculature divides into a dorsal **epimere** and a ventral **hypomere**.

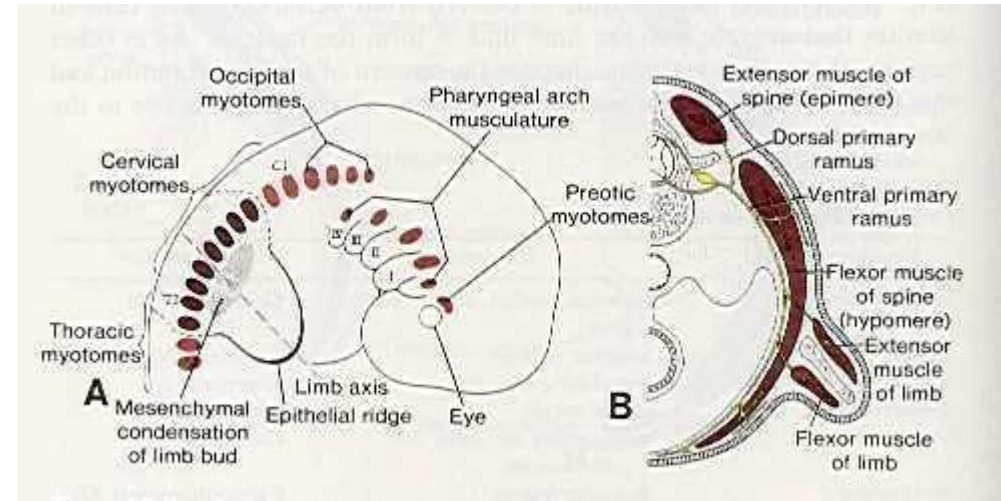


MUSCULAR SYSTEM DEVELOPMENT

SKELETAL MUSCLES

Muscles of the head - the extrinsic and intrinsic muscles of the tongue are derived from the occipital myotomes while extrinsic muscles of the eye are derived from the preoptic myotomes. The muscles of mastication, facial muscles of expression, muscles of pharynx and larynx are derived from the mesenchyme of the pharyngeal arches.

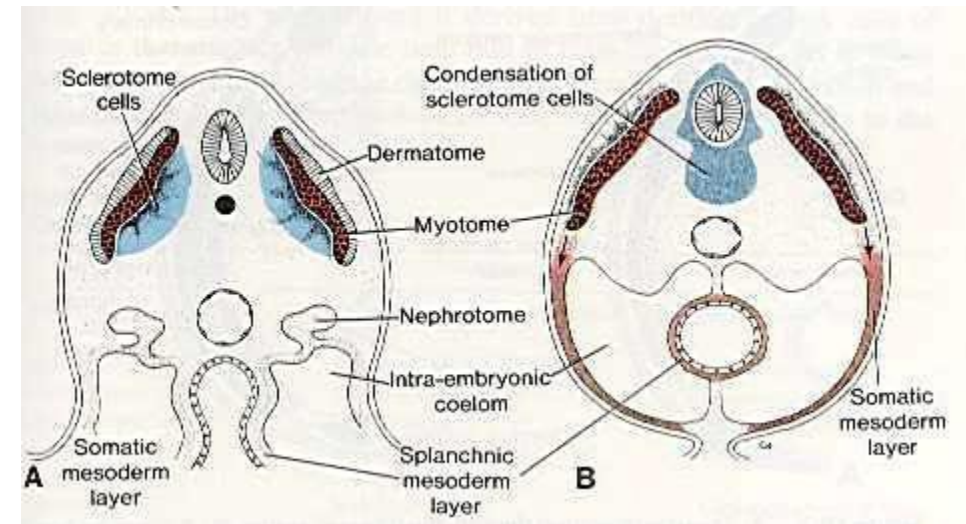
Limb muscles - in the seventh week, mesenchyme derived from the dermomyotome migrate into the limb bud; with elongation of the limb buds, the muscular tissue splits into flexor and extensor components.



MUSCULAR SYSTEM DEVELOPMENT

SMOOTH MUSCLES

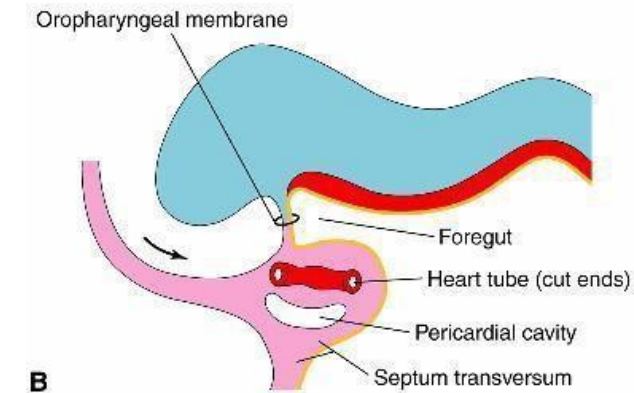
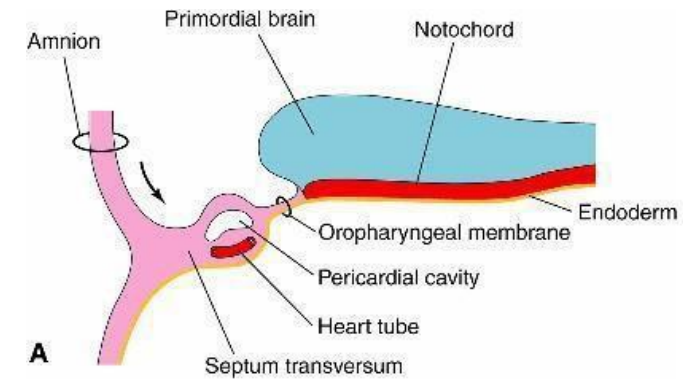
- Develop mainly from the splanchnic mesoderm layer surrounding the GI tract and its derivatives
- Blood vessels which develop in the limb buds, head, and body wall obtain their muscular coat from local mesenchyme; in fact, mesenchyme anywhere in the body is a potential source of smooth muscle tissue
- Smooth muscles of the iris, however, differentiate from the ectoderm of the optic cup
- Myoepithelial cells of the mammary glands and sweat glands originate from ectoderm



MUSCULAR SYSTEM DEVELOPMENT

CARDIAC MUSCLE

- Develops from the splanchnic mesoderm surrounding the endothelial heart tube
- The myoblasts adhere to one another by special attachments that later develop into intercalated discs



SKELETAL SYSTEM DEVELOPMENT

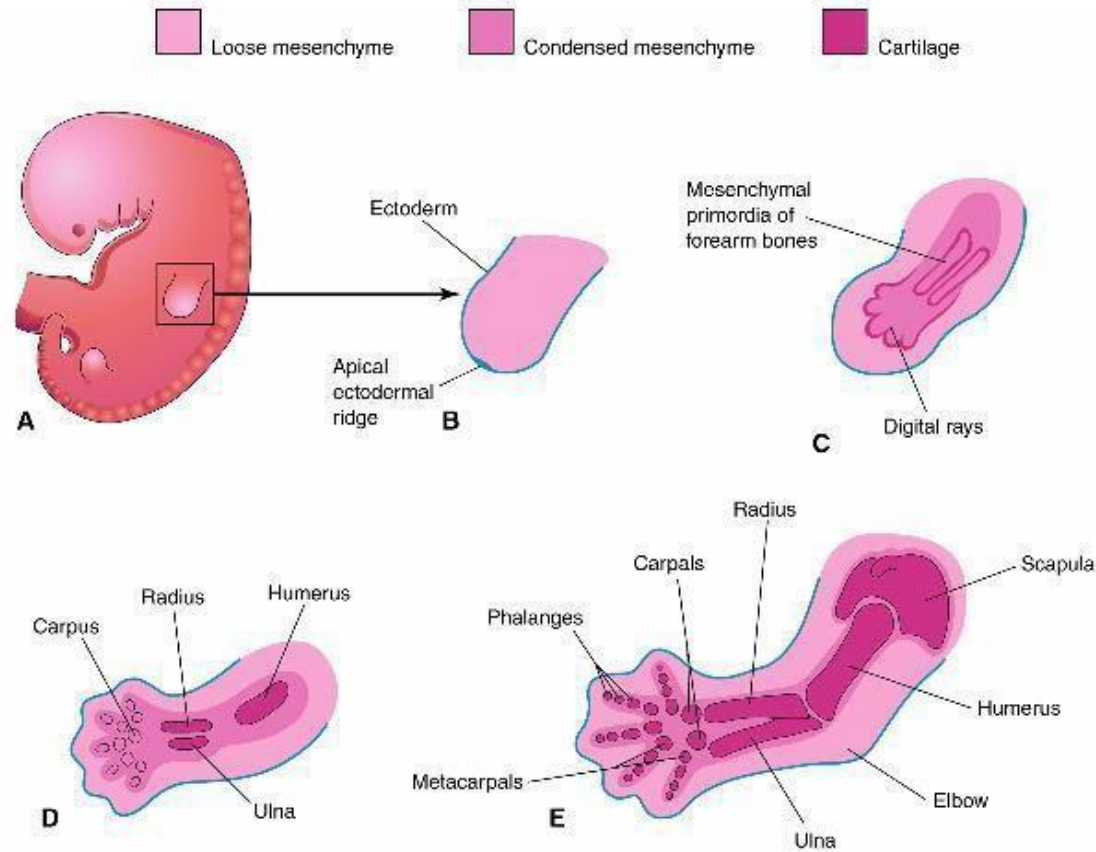
Develops from mesodermal and neural crest cells

- The paraxial mesoderm develop into segments called somites
- The somites differentiate into sclerotome and dermomyotome
- The sclerotomal cells form the vertebrae and ribs

The cranium develops from mesenchyme around the developing brain.

Appendicular bones develop from condensation of mesenchyme appearing in the limb buds during the fifth week of development

DEVELOPMENT OF SKELETAL SYSTEM



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References

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2. Before we are born: Essentials of embryology and birth defects by Moore, K. L., Persaud, T. V. N., & Torchia, M. G., 9th ed., 2016, Elsevier